

U-Net Adapter-based Universal Ultrasound Segmentation and Classification

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Abstract. Ultrasound imaging is a vital diagnostic modality, yet accurately analyzing images across diverse organs and pathologies remains a significant challenge due to limited generalization of existing task-specific models. In this work, we address this gap by developing a universal ultrasound image analysis model capable of performing multiple tasks, including classification and segmentation, across a broad range of anatomical regions and clinical conditions. Our approach leverages parameter-efficient adapters—small task-specific modules—allowing a single base model to be rapidly configured for new tasks at test time by adapting only a small subset of its parameters. This design not only facilitates multi-task capability but also reduces computational and data requirements for deployment in varied clinical settings. We developed and evaluated our model in the Universal UltraSound Image Challenge: Multi-Organ Classification and Segmentation (UUSIC), achieving a ranking among the top five teams during the validation phase. These results demonstrate the effectiveness and versatility of our approach for general-purpose ultrasound image analysis. **Team: cvhci; Team leader name: Zdravko Marinov; Team members: Zdravko Marinov, Rainer Stiefelhagen, Jens Kleesiek.**

Keywords: Ultrasound · Segmentation · Classification · Multi-task Learning · Universal Model

1 Introduction

Ultrasound imaging is an essential diagnostic tool in modern medicine due to its non-invasive nature, real-time capabilities, and cost-effectiveness. It is routinely employed for examining various organs, detecting tumors, and guiding interventional procedures [1, 12, 5]. Large-scale public datasets have been developed for organs such as the breast [19, 3], liver [2], kidney [14], thyroid [12], fetal head [5], and the heart [6], enabling rapid progress in computer-aided diagnosis (CAD) systems.

Despite these advances, developing models that generalize well across diverse anatomical regions and pathologies remains a major challenge. Existing

approaches often specialize in a single task or organ system [18], resulting in reduced adaptability to unseen data or different imaging contexts. The **Universal Ultrasound Image Challenge 2025 (UUSIC25)** addresses this limitation by introducing a benchmark that requires models to perform *both* classification and segmentation across multiple organs [7], using a mix of public and private datasets collected from several hospitals. The private test set ensures that successful models must exhibit robust generalization.

From a technical standpoint, building a universal ultrasound analysis model requires advanced machine learning and image processing techniques to extract meaningful features while maintaining high accuracy across tasks. However, training and storing separate models for each organ-task combination is inefficient and impractical for clinical deployment. To address this, our approach leverages *parameter-efficient adapters* in the form of a modulation mechanism: the model’s weights are factorized into a shared backbone and lightweight, task-specific modulation parameters, all stored within a single checkpoint. At inference time, the model can be prompted to activate the appropriate modulation, enabling it to perform a wide range of classification and segmentation tasks across multiple organs while maintaining the efficiency and simplicity of a universal model. This design enables a single base model to handle diverse classification and segmentation tasks with minimal additional computation and memory overhead, which is particularly important for real-time clinical applications where hardware resources may be limited [8]. This is important in the age of foundation models as models are becoming ever larger, and efficient inference would reduce the carbon footprint and ease the clinical deployment of these models [8], especially for interactive data annotation [9, 11, 4].

We developed and evaluated our model as part of the UUSIC25 challenge, where it achieved a top 10 spot in the ranking during the validation phase, confirming both the effectiveness and the efficiency of our approach for general-purpose ultrasound image analysis.

2 Method

2.1 Multi-Task Paradigm

We draw inspiration from the promptable multi-task framework proposed by Lin *et al.* [7], which demonstrated that a single model can be configured to perform both classification and segmentation across multiple ultrasound datasets via task-specific prompts. Our approach differs in two key aspects. First, instead of a transformer-based backbone, we employ modality-optimized architectures: a U-Net for segmentation tasks [13] and an EfficientNet encoder for classification tasks [16], both selected for their strong performance and efficiency in medical imaging. Second, our prompting mechanism is realized through *parameter-efficient adapters* configured via model modulation.

Formally, let θ_{shared} represent the shared backbone parameters and δ_t denote the modulation parameters associated with task t . At inference, the task-specific configuration θ_t is computed as:

$$\theta_t = \theta_{\text{shared}} \odot f(\delta_t), \quad (1)$$

where $f(\cdot)$ is a lightweight modulation function that adjusts the shared parameters in a task-specific manner. In practice, each δ_t can be interpreted as an element-wise scaling that selectively emphasizes or attenuates components of θ_{shared} to reproduce the behavior of a fully task-specific model. This allows us to store multiple task-specific configurations within a single checkpoint: although it appears as if we have a single universal model, each δ_t effectively encodes a separate trained model. At inference, the model can be “prompted” into different operational modes by applying the corresponding δ_t , and multiple tasks can even be combined by element-wise multiplication of their scaling parameters. Because element-wise scaling is fully reversible, individual task-specific behaviors can be recovered exactly, making this approach both flexible and parameter-efficient. This multi-task paradigm is also known as task-switching [15, 10, 17].

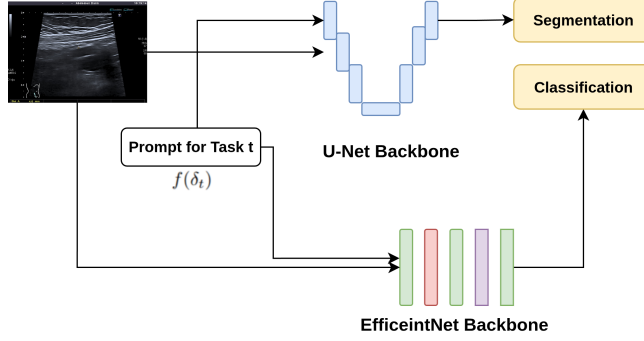


Fig. 1. Our model pipeline

2.2 Model Architectures

We use a U-Net backbone for all segmentation tasks [13] and an EfficientNet encoder for the classification tasks [16]. Similarly to Lin *et al.* [7], we have separate segmentation and classification branches in our framework. Our approach is illustrated in Fig. 1.

2.3 Loss Function

For segmentation tasks, we optimize a combined Dice and binary cross-entropy (BCE) loss:

$$\mathcal{L}_{\text{seg}} = 1 - \frac{2 \sum_i p_i g_i}{\sum_i p_i + \sum_i g_i} + \frac{-1}{N} \sum_{i=1}^N [g_i \log p_i + (1 - g_i) \log(1 - p_i)], \quad (2)$$

where p_i and g_i are the predicted and ground truth pixel values, respectively. For classification tasks, we use the standard BCE loss:

$$\mathcal{L}_{\text{cls}} = \frac{-1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)], \quad (3)$$

with y_i and $\hat{y}_i$ denoting the ground truth label and predicted probability.

We train classification tasks for 10 epochs, reflecting the smaller dataset size, and segmentation tasks for 50 epochs using both the public and private datasets provided by the UUSIC organizers. This unified yet flexible modulation-based framework allows us to maintain a single model file while delivering strong, specialized performance across multiple ultrasound analysis tasks.

3 Experimental Results

The results on the validation set are listed in Table 1. Although our approach ranked in the top 10 teams, we still see a lot of room for improvement for the tasks of Thyroid segmentation and all classification tasks. We believe that the classification task as a whole is more difficult due to the small sizes of these datasets, and since we only used the images as an input to the model. One way to improve the results is to use additional clinical data or to incorporate masks from segmentation models as an additional input.

Table 1. Performance metrics on the validation set

Task 1 (DSC)					Task 2 (Acc)			
Breast	Cardiac	Thyroid	Fetal Head	Kidney	Breast B/M	Breast Subtype	Appendix	Liver
0.81	0.82	0.57	0.93	0.88	0.77	0.61	0.63	0.74

4 Conclusion

In this work, we presented a universal ultrasound image analysis framework capable of performing both classification and segmentation across multiple organs using a single base model enhanced with parameter-efficient adapters. By modulating only a small set of task-specific parameters, our approach effectively encodes multiple specialized models within a single checkpoint, enabling flexible and reversible task switching without the need for multiple full models. Evaluations in the UUSIC25 challenge demonstrated competitive performance, particularly in breast, cardiac, fetal head, and kidney tasks, while highlighting areas for improvement in thyroid segmentation and classification tasks. Overall, our results indicate that lightweight, modulation-based multi-task models are a promising direction for scalable, general-purpose ultrasound analysis, facilitating efficient deployment in diverse clinical settings.

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